(i)

$$C = \begin{cases} ky, & X \leq y \\ ky + a(X - y), & X > y \end{cases}$$

$$\begin{aligned} \mathrm{E}(C) &= \int_0^y ky\lambda e^{-\lambda x}\,dx + \int_y^\infty \left[ky + a(x-y)\right]\lambda e^{-\lambda x}\,dx \\ &= ky\int_0^\infty \lambda e^{-\lambda x}\,dx + a\int_y^\infty (x-y)\lambda e^{-\lambda x}\,dx \\ &= ky + a\left[-(x-y)e^{-\lambda x} - \frac{1}{\lambda}e^{-\lambda x}\right]_y^\infty \\ &= ky + \frac{a}{\lambda}e^{-\lambda y} \end{aligned}$$

$$\frac{d}{dy}\mathrm{E}(C) = k - ae^{-\lambda y} \quad \frac{d^2}{dy^2}\mathrm{E}(C) = a\lambda e^{-\lambda y} > 0$$

$$k - ae^{-\lambda y} = 0 \quad \Longrightarrow \quad y = \frac{1}{\lambda}\ln\left(\frac{a}{k}\right)$$

$$\boxed{\therefore y = \begin{cases} 0, & \frac{a}{k} \leq 1 \\ \frac{1}{\lambda}\ln\left(\frac{a}{k}\right), & \frac{a}{k} > 1 \end{cases}}$$

(ii) Let $u = x - y$

$$\begin{aligned} \mathrm{E}(C^2) &= k^2y^2 + \int_y^\infty \left[2aky(x-y) + a^2(x-y)^2\right]\lambda e^{-\lambda x}\,dx \\ &= k^2y^2 + e^{-\lambda y}\int_0^\infty \left(2akyu + a^2u^2\right)\lambda e^{-\lambda u}\,du \\ &= k^2y^2 + e^{-\lambda y}\left(2aky\int_0^\infty u\lambda e^{-\lambda u}\,du + a^2\int_0^\infty u^2\lambda e^{-\lambda u}\,du\right) \\ &= k^2y^2 + e^{-\lambda y}\left(\frac{2aky}{\lambda} + a^2\left[-u^2e^{-\lambda u}\right]_0^\infty + 2a^2\int_0^\infty ue^{-\lambda u}\,du\right) \\ &= k^2y^2 + \left(\frac{2aky}{\lambda} + \frac{2a^2}{\lambda^2}\right)e^{-\lambda y} \end{aligned}$$

$$\begin{aligned} \mathrm{Var}(C) &= \mathrm{E}(C^2) - (\mathrm{E}(C))^2 \\ &= k^2y^2 + \left(\frac{2aky}{\lambda} + \frac{2a^2}{\lambda^2}\right)e^{-\lambda y} - \left(ky + \frac{a}{\lambda}e^{-\lambda y}\right)^2 \\ &= \boxed{\frac{a^2}{\lambda^2}\left(2e^{-\lambda y} - e^{-2\lambda y}\right)} \end{aligned}$$

$$\frac{d}{dy}\text{Var}(C) = \frac{2a^2}{\lambda}e^{-\lambda y}\left(e^{-\lambda y} - 1\right)$$

For $y > 0$, $\quad e^{-\lambda y} < 1 \quad \Longrightarrow \quad \frac{d}{dy}\text{Var}(C) < 0$

As y increase, Var(y) decreases

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: June 20, 2026